

Technical Design Document (TDD)

System Architecture, API Specifications & Kinship Engine Specs

Version: 1.0.0 Date: July 2026 Status: Approved Architecture Stack: Full-Stack Web Platform

1. System Tech Stack & Layered Architecture

Layer	Selected Technology	Technical Justification & Function
Frontend Framework	Next.js 14+ (React + TypeScript)	Server-Side Rendering (SSR) for initial load, robust client state management, and end-to-end type safety.
Canvas Engine	React Flow (v11+)	High-performance rendering engine for infinite canvases, supporting custom node components, viewport zoom/pan, and virtualization.
UI & Styling	Tailwind CSS + Shadcn/ui	Responsive, accessible component library optimized for lightweight UI bundles.
Backend API	Node.js (NestJS / Next.js App Router)	Modular architecture for controller endpoints, deduplication algorithms, and kinship graph resolution.
Realtime & Auth	Supabase Auth + Realtime Engine	JWT-based authentication and WebSocket subscriptions for instant canvas updates upon steward approval.
Database & ORM	PostgreSQL + Drizzle ORM	Native support for Recursive CTEs, type-safe SQL queries, and lightweight query execution overhead.

2. Canvas Rendering & Performance Optimization

The core presentation UI relies on **React Flow** with custom **PersonNode** components. To maintain 60 FPS performance when panning large trees:

- Bounding-Box Fetching (Viewport Virtualization):** The client requests nodes strictly within the visible viewport bounding box (**xMin, yMin, xMax, yMax**).
- Node Memoization:** Wrap custom node components in **React.memo** to prevent unnecessary re-renders during canvas dragging.

3. RESTful API Specification

Method	Endpoint Path	Description & Function	Auth & Role
GET	/api/v1/tree/canvas	Fetch nodes and edges for the viewport bounding box.	User Authenticated
POST	/api/v1/persons	Create a new individual node and direct relationship.	User Authenticated
POST	/api/v1/dedup/check	Real-time background search for matching duplicate persons.	Debounced Realtime

Method	Endpoint Path	Description & Function	Auth & Role
POST	/api/v1/claim/request	Submit a profile claim request ("This is Me").	User Authenticated
PUT	/api/v1/review/approve	Approve pending relation link (PENDING $\rightarrow$ VERIFIED).	Branch Steward / Admin
POST	/api/v1/review/merge	Merge two duplicate person nodes into a single record.	Branch Steward / Admin

4. Kinship Engine & Recursive SQL Queries

Ancestry traversal is executed directly in PostgreSQL using **Recursive Common Table Expressions (CTE)**:

```
WITH RECURSIVE Ancestors AS (  
  -- Anchor Member: Direct Verified Parents  
  SELECT r.person_id, r.related_person_id, r.relationship_type, 1 AS depth  
  FROM relationships r  
  WHERE r.person_id = :target_person_id  
    AND r.relationship_type = 'PARENT'  
    AND r.status = 'VERIFIED'  
  
  UNION ALL  
  
  -- Recursive Member: Traverse Upward to Ancestors  
  SELECT r.person_id, r.related_person_id, r.relationship_type, a.depth + 1  
  FROM relationships r  
  INNER JOIN Ancestors a ON r.person_id = a.related_person_id  
  WHERE r.relationship_type = 'PARENT'  
    AND r.status = 'VERIFIED'  
)  
SELECT a.depth, p.id, p.first_name, p.family_name, p.gender  
FROM Ancestors a  
JOIN persons p ON p.id = a.related_person_id  
ORDER BY a.depth ASC;
```

5. Real-time Deduplication Scoring Formula

The deduplication engine calculates a composite **Matching Score**:

Matching Score = (Sim_Name × 0.50) + (Sim_Context × 0.30) + (Sim_BirthYear × 0.20)

- **Name Similarity (Sim_Name):** Modified Levenshtein Distance handling phonetic variations and Arabic name normalization.
- **Context Similarity (Sim_Context):** Evaluates overlap in parent, grandparent, and family branch lineage.
- **Birth Year Proximity (Sim_BirthYear):** 1.0 for exact match, 0.5 for difference ≤ 3 years, 0 otherwise.